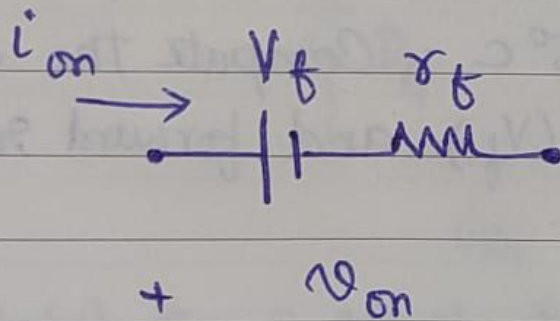


Assignment - 1

1. Power devices can typically be represented as a voltage source in series with a resistance in their ON state as shown.

V_f and r_f represent the forward voltage drop and forward resistance of the device, respectively.



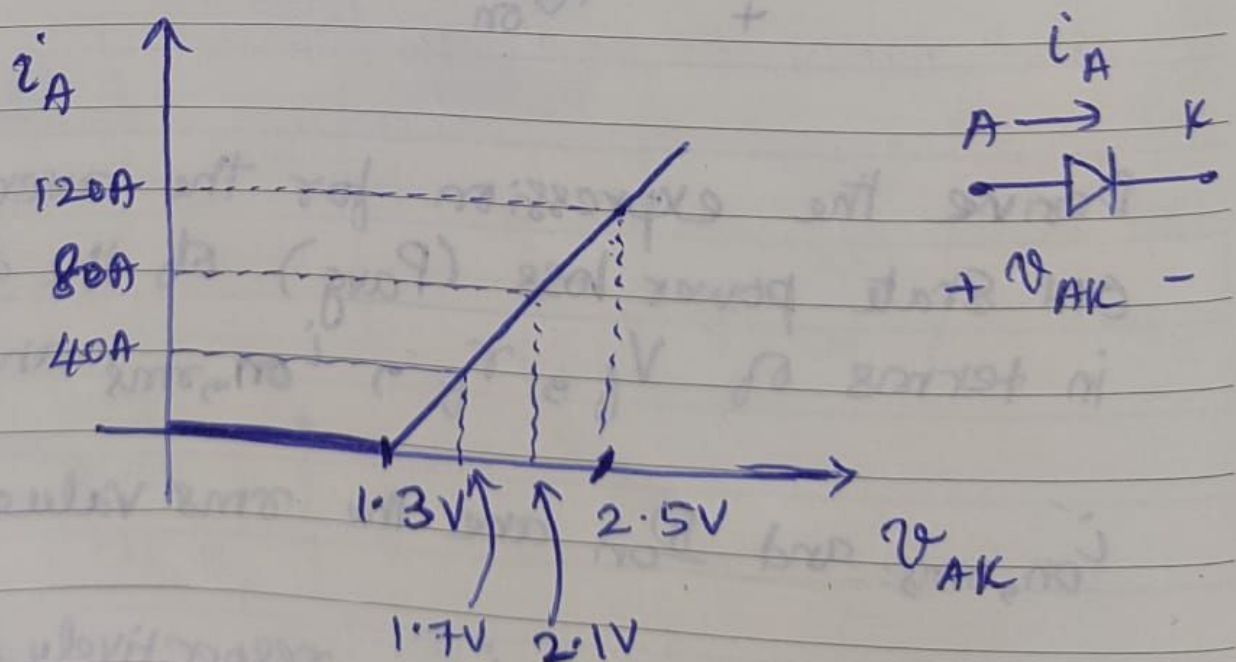
Derive the expression for the average ON state power loss (P_{avg}) of the device in terms of V_f , r_f , $i_{on,rms}$ and D_{on} .

$i_{on,rms}$ and D_{on} are the rms value and average value of i_{on} , respectively.

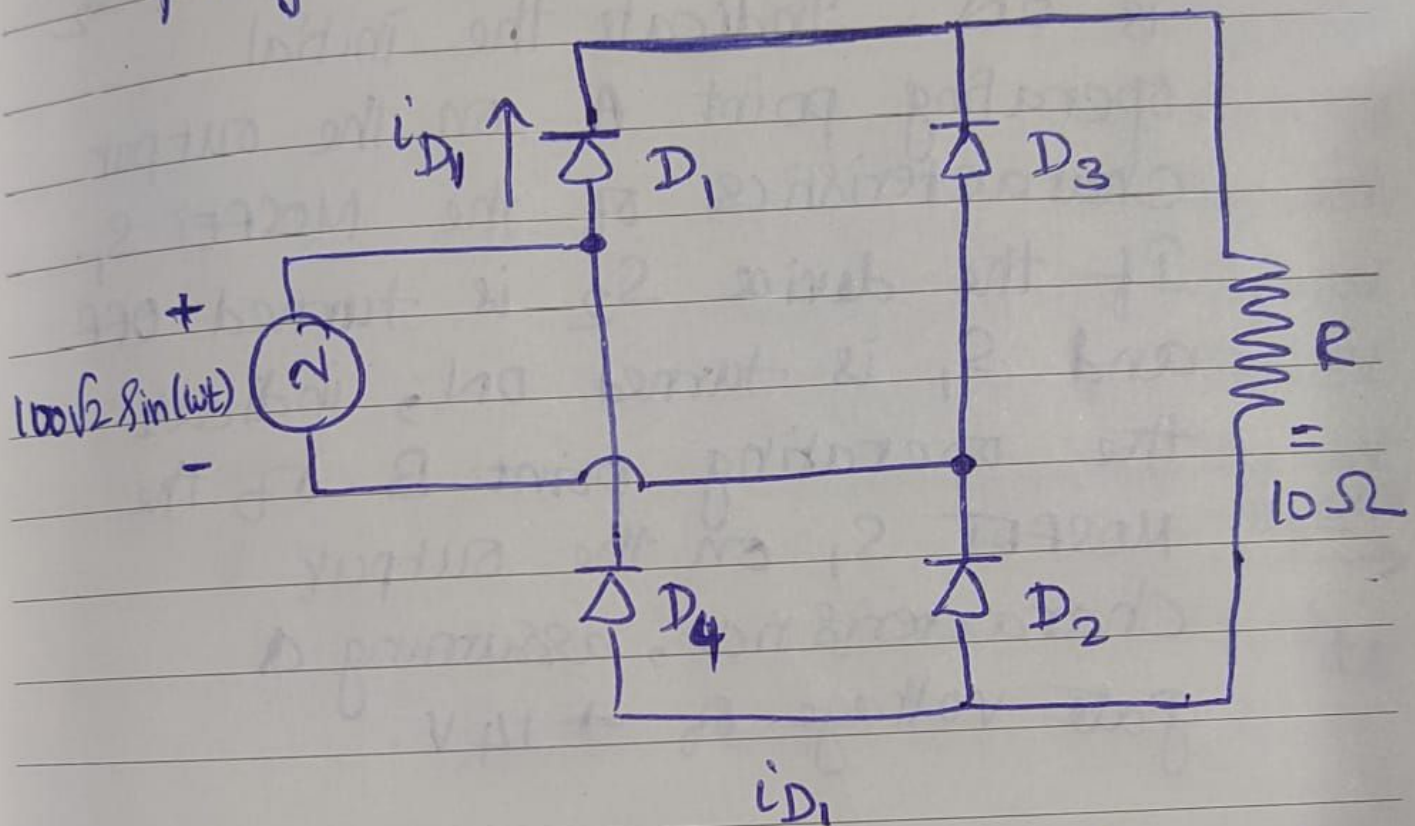
Hint: $P_{avg} = \frac{1}{T} \int_0^T p_{on}(t) dt$

$\Rightarrow P_{avg} = \frac{1}{T} \int_0^T v_{on}(t) \cdot i_{on}(t) dt$

2. The figure roughly represents the $v-i$ characteristics of the power diode Infineon IDWD120E120D7, a 1200V/120A device at 25°C. Compute the forward voltage drop (V_f) and forward resistance (r_f) of the device.



3. A full bridge diode rectifier feeding a purely resistive load is shown.



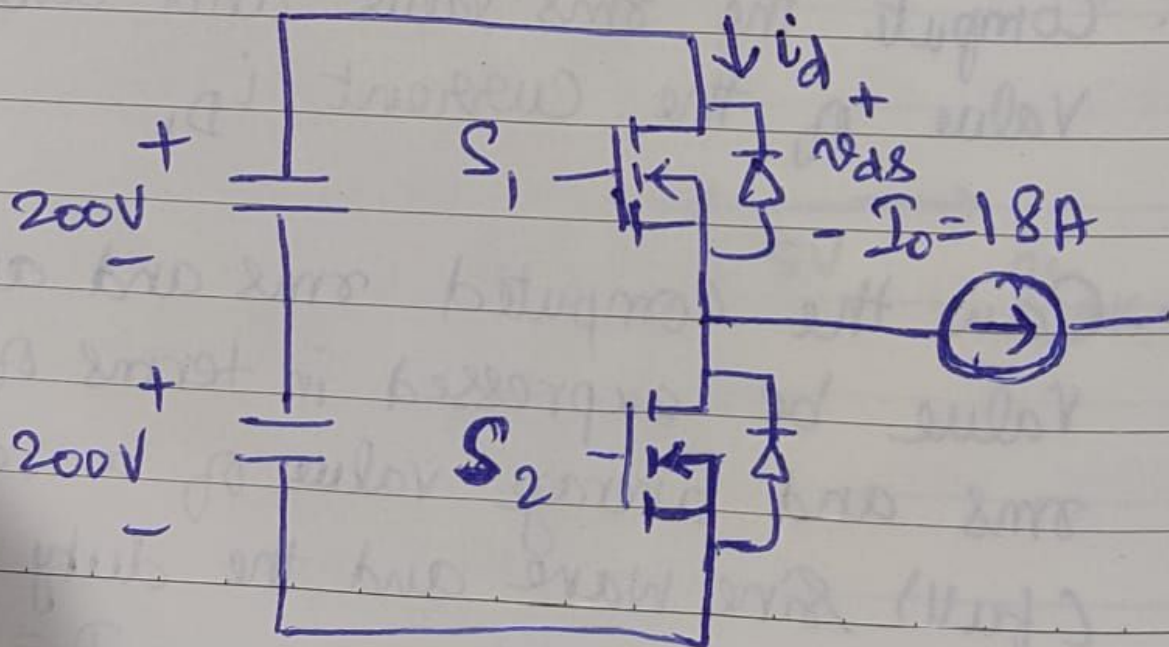
a. Sketch the current i_{D1} through the Diode D_1 .

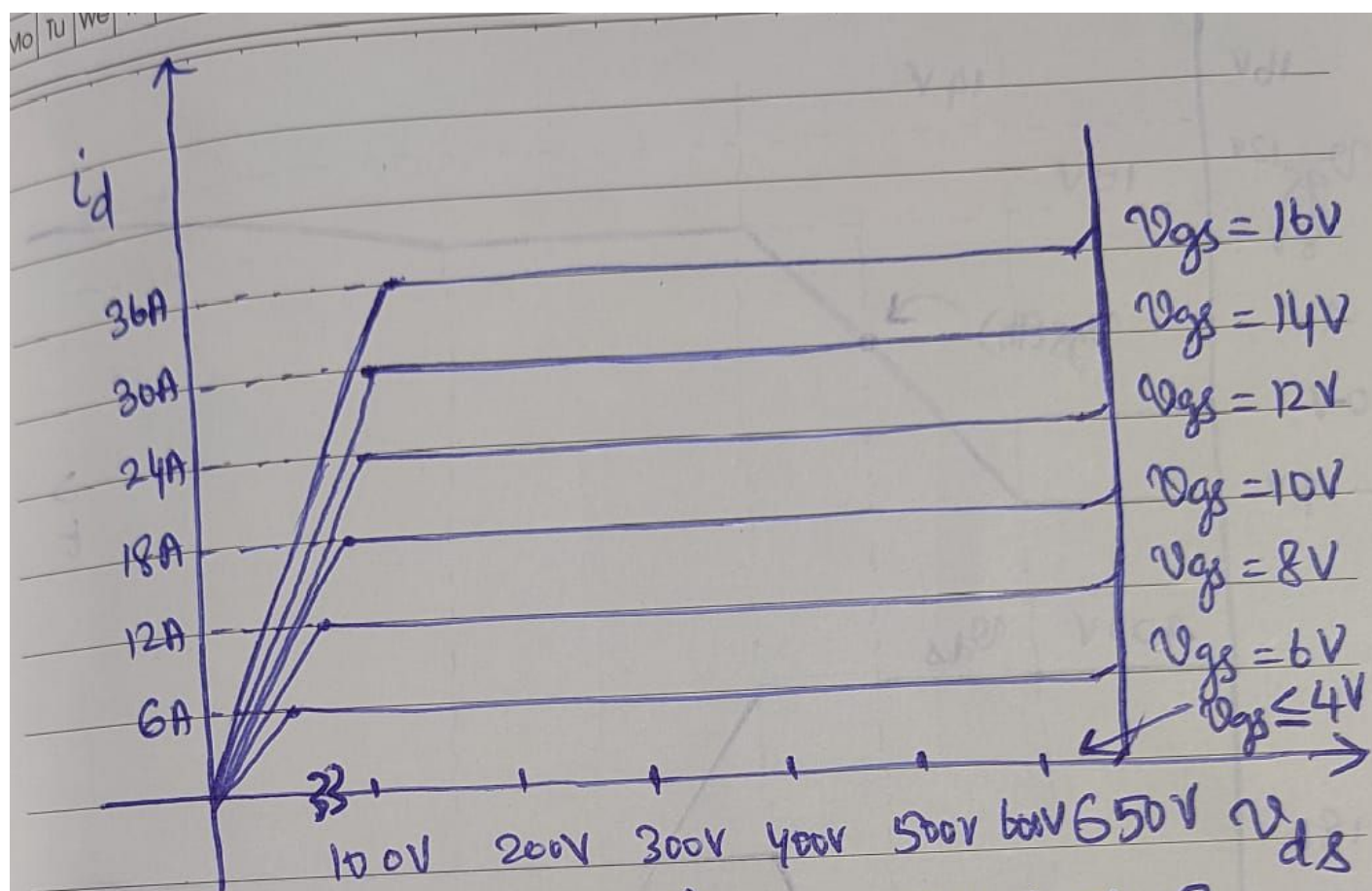
b. Compute the rms value and average value of the current i_{D1} .

c. Can the computed rms and average value be expressed in terms of the rms and average value of a rectified (full) sine wave and the duty ratio $D = 1/2$?

4. In the circuit below, initially, MOSFET S_1 is OFF and MOSFET S_2 is ON. Indicate the initial operating point A on the output characteristics of the MOSFET S_1 . If the device S_2 is turned OFF and S_1 is turned ON, indicate the operating point B of the MOSFET S_1 on the output characteristics, assuming a gate voltage of $+14V$.

Also, identify the operating region (cut-off, ohmic, active or breakdown) of the device at A and B.

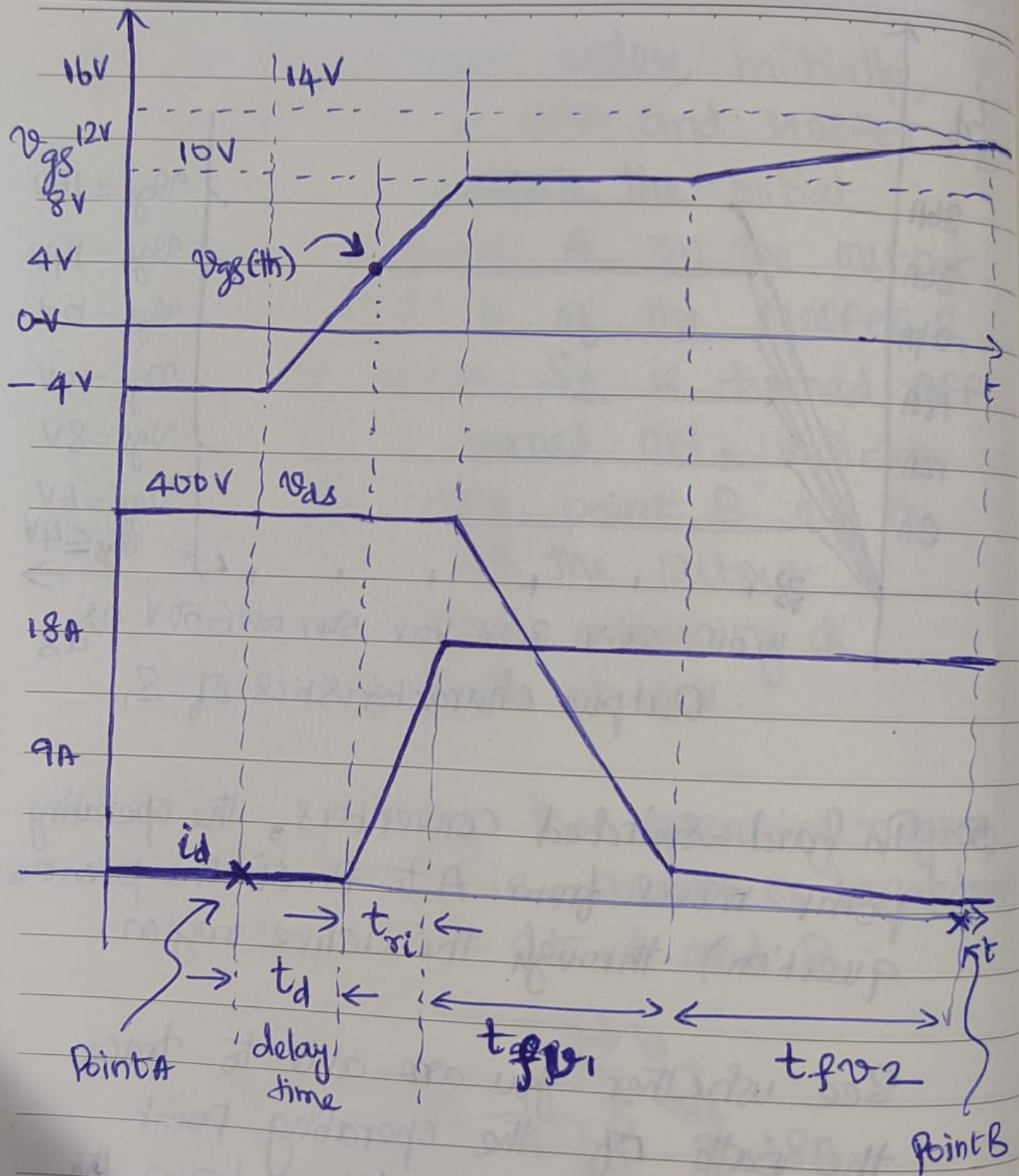




Output characteristics of S_1 .

5. In hard-switched converters, the operating point moves from A to B (in the previous question) through the active region.

See whether you are able to trace the path of the operating point when the device transitions from the OFF state to the ON state using the time plots given below.



6. In the previous question, you can observe that the switching loss happens during t_{ri} , t_{fvi} , t_{fv2} intervals.

Assuming the voltage V_{ds} to be zero at the end of t_{fvi} , sketch the instantaneous power loss during durations t_{ri} and t_{fvi} .

Observe where the peak power loss happens and how the power loss varies with time during switching.